

QMFM model specification and parameters calibration
Nov. 2023 round

```
% -----
% ----- Transition variables -----
% -----
!transition_variables
```

% ----- Components of real GDP -----

% ----- Private consumption -----

```
"Consumption gap, %" l_cons_gap
"Exptd. cons. gap, %" e_l_cons_gap
```

```
"Consumption, 100*log" l_cons
```

```
"Consumption, ann. QQ %" dl_cons
```

```
"Consumption, YY %" d4l_cons
```

```
"Consumption tnd., 100*log" l_cons_tnd
```

```
"Consumption tnd., ann. QQ %" dl_cons_tnd
```

```
"Consumption tnd., YY %" d4l_cons_tnd
```

% ----- Private investment -----

```
"Investment gap, %" l_inv_gap
```

```
"Exptd. investment gap, %" e_l_inv_gap
```

```
"Investment, 100*log" l_inv
```

```
"Investment, ann. QQ %" dl_inv
```

```
"Investment, YY %" d4l_inv
```

```
"Investment tnd., 100*log" l_inv_tnd
```

```
"Investment tnd., ann. QQ %" dl_inv_tnd
```

```
"Investment tnd., YY %" d4l_inv_tnd
```

% ----- Government demand for G&S -----

```
"Gov. demand gap, %" l_gdem_gap
```

```
"Gov. demand, 100*log" l_gdem
```

```
"Gov. demand, ann. QQ %" dl_gdem
```

```
"Gov. demand, YY %" d4l_gdem
```

```
"Gov. demand, tnd., 100*log" l_gdem_tnd
```

```
"Gov. demand, tnd., ann. QQ %" dl_gdem_tnd
```

```
"Gov. demand, tnd., YY %" d4l_gdem_tnd
```

% ----- Exports of G&S -----

```
"Export gap, %" l_exp_gap
```

```
"Exptd. export gap, %" e_l_exp_gap
```

```
"Export RMCI, %" rmc_i_exp
```

```
"Export, 100*log" l_exp
```

```
"Export, ann. QQ %" dl_exp
```

```
"Export, YY %" d4l_exp
```

```
"Export tnd., 100*log" l_exp_tnd
```

```
"Export tnd., ann. QQ %" dl_exp_tnd
```

```
"Export tnd., YY %" d4l_exp_tnd
```

% ----- Imports of G&S -----

```
"Import gap, %" l_imp_gap
```

```
"Import, 100*log" l_imp
```

```
"Import, ann. QQ %" dl_imp
```

```
"Import, YY %" d4l_imp
```

```
"Import tnd., 100*log" l_imp_tnd
```

```
"Import tnd., ann. QQ %" dl_imp_tnd
```

```
"Import tnd., YY %" d4l_imp_tnd
```

% ----- Fiscal policy -----

```
"Fiscal impulse, % of GDP" fisc_imp
```

```
"Budget deficit, % of GDP" def_y
```

```
"Struct. deficit, % of GDP" def_y_str
```

```
"Cyc. deficit, % of GDP" def_y_cyc
```

```
"Discr. deficit, % of GDP" def_y_discr
```

```
"Govt. demand (G&S), % GDP" gdem_y
```

```
"Struct. govt. demand (G&S), % GDP" gdem_y_str
```

```
"Cyc. govt. demand (G&S), % GDP" gdem_y_cyc
```

```
"Discr. govt. demand (G&S), % GDP" gdem_y_discr
```

```
"Other govt. exp., % to GDP" oexp_y
```

```
"Cyc. other govt. exp., % to GDP" oexp_y_cyc
```

```
"Struct. other govt. exp., % to GDP" oexp_y_str
```

```
"Discr. other govt. exp., % to GDP" oexp_y_discr
```

```
"Govt. rev., % of GDP" grev_y
```

```
"Struct. govt. rev., % of GDP" grev_y_str
```

```
"Cyc. govt. rev., % of GDP" grev_y_cyc
```

```
"Discr. govt. rev., % of GDP" grev_y_discr
```

```
"Fiscal grants, % of GDP" grants_y
```

% ----- GDP -----

```
"Output gap, %" l_y_gap
```

```
"Exptd. output gap, %" e_l_y_gap
```

```
"GDP, 100*log" l_y
```

```
"GDP, ann. QQ %" dl_y
```

```
"GDP, YY %" d4l_y
```

```
"GDP tnd., 100*log" l_y_tnd
```

```
"GDP tnd., ann. QQ %" dl_y_tnd
```

```
"GDP tnd., YY %" d4l_y_tnd
```

% ----- Agricultural output -----

```
"Agric. output gap, %" l_y_agr_gap
```

```
"Exptd. agric. output gap, %" e_l_y_agr_gap
```

```
"Agric. GDP, 100*log" l_y_agr
```

```
"Agric. GDP, ann. QQ %" dl_y_agr
```

```
"Agric. GDP, YY %" d4l_y_agr
```

```
"Agric. GDP tnd., 100*log" l_y_agr_tnd
```

```
"Agric. GDP tnd., ann. QQ %" dl_y_agr_tnd
```

```
"Agric. GDP tnd., YY %" d4l_y_agr_tnd
```

```

% -----
% ----- Prices -----
% ----- Headline CPI -----

"Headline CPI, 100*log" l_cpi
"Headline CPI, ann. QQ %" dl_cpi
"Exptd. Headline CPI, ann. QQ %" e_dl_cpi

"Headline CPI, YY %" d4l_cpi
% ----- Core CPI -----
"Core CPI, 100*log" l_cpi_core
"Core CPI, ann. QQ %" dl_cpi_core
"Exptd. Core CPI, ann. QQ %" e_dl_cpi_core
"Core CPI, YY %" d4l_cpi_core

% ----- Food CPI -----
"Food CPI, 100*log" l_cpi_food
"Food CPI, ann. QQ %" dl_cpi_food
"Exptd. Food CPI, ann. QQ %" e_dl_cpi_food
"Food CPI, YY %" d4l_cpi_food

% ----- Energy CPI -----
"Energy CPI, 100*log" l_cpi_ener
"Energy CPI, ann. QQ %" dl_cpi_ener
"Exptd. energy CPI, ann. QQ %" e_dl_cpi_ener
"Energy CPI, YY %" d4l_cpi_ener

% -----
% ----- Relative prices -----
% ----- Real marginal cost gap -----
"RMC, %" rmc

% ----- Core/headline relative prices -----
"Core rel. price, 100*log" l_rp_cpi_core
"Core rel. price, ann. QQ %" dl_rp_cpi_core
"Core rel. price, YY %" d4l_rp_cpi_core
"Core rel. price gap, %" l_rp_cpi_core_gap
"Core rel. price tnd., 100*log" l_rp_cpi_core_tnd
"Core rel. price tnd., ann. QQ %" dl_rp_cpi_core_tnd
"Core rel. price tnd., YY %" d4l_rp_cpi_core_tnd
"Dir. ext. eff. in core, ann. QQ %" dl_cpi_core_direct

% ----- Food/headline relative prices -----

"Food rel. price, 100*log" l_rp_cpi_food
"Food rel. price, ann. QQ %" dl_rp_cpi_food
"Food rel. price, YY %" d4l_rp_cpi_food
"Food rel. price gap, %" l_rp_cpi_food_gap
"Food rel. price tnd., 100*log" l_rp_cpi_food_tnd
"Food rel. price tnd., ann. QQ %" dl_rp_cpi_food_tnd
"Food rel. price tnd., YY %" d4l_rp_cpi_food_tnd
"Dir. ext. eff. in food, ann. QQ %" dl_cpi_food_direct

% ----- Energy/headline relative prices -----
"Energy rel. price, 100*log" l_rp_cpi_ener
"Energy rel. price, ann. QQ %" dl_rp_cpi_ener
"Energy rel. price, YY %" d4l_rp_cpi_ener
"Energy rel. price gap, %" l_rp_cpi_ener_gap
"Energy rel. price tnd., 100*log" l_rp_cpi_ener_tnd
"Energy rel. price tnd., ann. QQ %" dl_rp_cpi_ener_tnd
"Energy rel. price tnd., YY %" d4l_rp_cpi_ener_tnd
"Dir. ext. eff. in energy, ann. QQ %" dl_cpi_ener_direct

```

```

% -----
% ----- Monetary policy -----
% ----- Nominal interest rate -----
"Policy rate (IB rate used), %" i
"Policy rate tnd (IB rate used), %" i_tnd

% ----- Real interest rate -----
"Real interest (policy) rate, %" r
"RIR (policy) gap, %" r_gap
"Exptd. RIR (policy) gap, %" e4_r_gap
"Exptd. RIR (lending) gap, 4-Q ahead av., %" r4_gap

"RIR (policy) trend, %" r_tnd

% ----- Lending premium -----

"Lending premium, %" prem_d
"Lending premium gap, %" prem_d_gap

% ----- Inflation target -----
"Inflation target, YY %" d4l_cpi_tar
"Headline CPI dev, 4-Q ahead YY %" d4l_cpi_dev
"Impl. core infl. target, YY %" d4l_cpi_core_tar
"Impl. food infl. target, YY %" d4l_cpi_food_tar
"Impl. ener. infl. target, YY %" d4l_cpi_ener_tar

% ----- Real monetary condition index -----
"RMCI cons., %" rmci_cons
"RMCI invest., %" rmci_inv

% -----
% ----- Exchange rate -----
% ----- Nominal exchange rate -----
"Exchg. rate, 100*log" l_s
"Exchg. rate, ann. QQ %" dl_s
"Exchg. rate, YY %" d4l_s
"Exptd. exchg. rate, 100*log" e_l_s
"Exchg rate target, ann. QQ %" dl_s_tar

% ----- Sovereign risk premium -----
"Risk premium, %" prem

% ----- Real exchange rate -----
"Real exchg. rate, 100*log" l_z

"Real exchg. rate, ann. QQ %" dl_z
"Real exchg. rate, YY %" d4l_z
"Exptd. RER, ann. QQ %" e_dl_z
"Real exchg. rate trend, 100*log" l_z_tnd
"Real exchg. rate trend, ann. QQ %" dl_z_tnd
"Real exchg. rate trend, YY %" d4l_z_tnd
"Exp. RER trend., ann. QQ %" e_dl_z_tnd
"Real exchg. rate gap, %" l_z_gap

% ----- Money demand -----
"Money, 100*log" l_md
"Real money, 100*log" l_rmd
"Real money, ann. QQ %" dl_rmd
"Real money, YY %" d4l_rmd
"Change of velocity, ann. QQ %" dl_v

```

```

% -----
% ----- External variables -----
% ----- External demand -----

"Foreign output gap, %" l_ystar_gap

% ----- Foreign prices -----
"Foreign CPI, 100*log" l_cpistar
"Foreign CPI, ann. QQ %" dl_cpistar
"Foreign CPI, YY %" d4l_cpistar
"Exptd. foreign CPI, 100*log" e_dl_cpistar

% ----- Foreign interest rate -----
"Foreign interest rate, %" istar
"Foreign real interest rate trend, %" rstar_tnd

% ----- World food prices -----
"Foreign food price, 100*log" l_foodstar
"Foreign food price, ann. QQ %" dl_foodstar
"Foreign food price, YY %" d4l_foodstar

"Foreign rel. food price, 100*log" l_rp_foodstar
"Foreign rel. food price, ann. QQ %" dl_rp_foodstar
"Foreign rel. food price, YY %" d4l_rp_foodstar
"Foreign rel. food price gap, %" l_rp_foodstar_gap
"Foreign rel. food price tnd., 100*log" l_rp_foodstar_tnd
"Foreign rel. food price tnd., ann. QQ %" dl_rp_foodstar_tnd
"Foreign rel. food price tnd., YY %" d4l_rp_foodstar_tnd

% ----- World energy prices -----
"Foreign ener. price, 100*log" l_enerstar
"Foreign ener. price, ann. QQ %" dl_enerstar
"Foreign ener. price, YY %" d4l_enerstar
"Foreign rel. ener. price, 100*log" l_rp_enerstar
"Foreign rel. ener. price, ann. QQ %" dl_rp_enerstar
"Foreign rel. ener. price, YY %" d4l_rp_enerstar
"Foreign rel. ener. price gap, %" l_rp_enerstar_gap
"Foreign rel. ener. price tnd., 100*log" l_rp_enerstar_tnd
"Foreign rel. ener. price tnd., ann. QQ %" dl_rp_enerstar_tnd
"Foreign rel. ener. price tnd., YY %" d4l_rp_enerstar_tnd

```

```

% -----
% ----- Transition shocks -----
% -----
!transition_shocks

"Fiscal grants, % of GDP" shock_grants_y(σ=2.5)
"Gdem. cyc. shock, % of GDP" shock_gdem_y_str(σ=0.075)
"Gdem. str. shock, % of GDP" shock_gdem_y_cyc(σ=0)
"Gdem. discr. shock, % of GDP" shock_gdem_y_discr(σ=1.7321)

"Gov exp. cyc. shock, % of GDP" shock_oexp_y_str(σ=0.1732)
"Gov exp. str. shock, % of GDP" shock_oexp_y_cyc(σ=0)
"Gov exp. discr. shock, % of GDP" shock_oexp_y_discr(σ=1.7321)
"Gov rev. str. shock, % of GDP" shock_grev_y_str(σ=0.1)
"Gov rev. cyc. shock, % of GDP" shock_grev_y_cyc(σ=0)
"Gov rev. discr. shock, % of GDP" shock_grev_y_discr(σ=1.7321)

"Cons. gap shock, %" shock_l_cons_gap(σ=2.5)
"Inv. gap shock, %" shock_l_inv_gap(σ=6)
"Export gap shock, %" shock_l_exp_gap(σ=6)
"Import gap shock, %" shock_l_imp_gap(σ=0.5)
"Output gap shock, %" shock_l_y_gap(σ=0.0001)
"Agric. output gap shock, %" shock_l_y_agr_gap(σ=0.3)

"Core infl. shock, ann. QQ %" shock_dl_cpi_core(σ=2)
"Food infl. shock, ann. QQ %" shock_dl_cpi_food(σ=9)
"Energy infl. shock, ann. QQ %" shock_dl_cpi_ener(σ=7.1)
"CPI discr. shock, 100*log" shock_l_cpi(σ=0.0001)

"Policy rate shock, %" shock_i(σ=1)
"Lending premium shock, %" shock_prem_d_gap(σ=0.8)
"Exchng. rate shock, 100*log" shock_l_s(σ=0.35)
"Exchng. rate target shock, ann. QQ %" shock_dl_s_tar(σ=0.3)
"Real money dem. shock, ann. QQ %" shock_dl_rmd(σ=1)
"Chng. of vel. shock, ann. QQ %" shock_dl_v(σ=0.1)

"Cons. tnd. shock, ann. QQ %" shock_dl_cons_tnd(σ=0.25)
"Inv. tnd. shock, ann. QQ %" shock_dl_inv_tnd(σ=0.5)
"Export shock, ann. QQ %" shock_dl_exp_tnd(σ=0.75)
"Import tnd. shock, ann. QQ %" shock_dl_imp_tnd(σ=0.25)

"CPI target shock, YY %" shock_d4l_cpi_tar(σ=0.1)
"Food rel. price. tnd. shock, ann. QQ %" shock_dl_rp_cpi_food_tnd(σ=0.5)
"Energy rel. price. tnd. shock, ann. QQ %" shock_dl_rp_cpi_ener_tnd(σ=0.15)
"RER tnd. shock, ann. QQ %" shock_dl_z_tnd(σ=0.1)
"Premium shock, %" shock_prem(σ=0.1)
"Agric. output tnd. shock, %" shock_dl_y_agr_tnd(σ=0.06)

"Foreign output gap shock, %" shock_l_ystar_gap(σ=0.45)
"Foreign CPI shock, ann. QQ %" shock_dl_cpistar(σ=3.8)
"Foreign intr. rate shock, %" shock_istar(σ=0.45)
"Foreign rel. food price gap shock, %" shock_l_rp_foodstar_gap(σ=5.3)
"Foreign rel. ener. price gap shock, %" shock_l_rp_enerstar_gap(σ=13.9)
"Foreign rel intr. rate rtd. shock, %" shock_rstar_tnd(σ=0.5)
"Foreign rel. food price tnd. shock, ann. QQ %"
shock_dl_rp_foodstar_tnd(σ=0.5)
"Foreign rel. ener. price tnd. shock, ann. QQ %"
shock_dl_rp_enerstar_tnd(σ=2.5)

```

```

% -----
% ----- Parameters -----
% -----

!parameters
% -----
% ----- Steady states -----

"steady state govt rev in % of GDP" ss_grev_y_str<21>
"steady state govt other exp in % of GDP" ss_oexp_y_str<6>
"steady state govt other exp in % of GDP" ss_gdem_y_str<26>

"SS of borrowing ratio, % of GDP" ss_bor_str<6>
"SS of FCY debt share" ss_debt_fcy_rat<0.846154>
"SS of fiscal grants ratio, % GDP" ss_grants_y<5>

"SS of FCY gov. debt. intr. prem., %" ss_prem_debt_fcy<NaN>
"SS of LCY gov. debt. intr. prem., %" ss_prem_debt_lcy<NaN>
"SS of GDP growth, ann. QQ %" ss_dl_y_tnd<7.23207>
"SS of agric.GDP growth, ann. QQ %" ss_dl_y_agr_tnd<7.23207>

"CPI target, YY %" ss_d4l_cpi_tar<4.87902>
"SS of food rel. price, ann. QQ %"
ss_dl_rp_cpi_food_tnd<1.98026>
"SS of ener. rel. price, ann. QQ %" ss_dl_rp_cpi_ener_tnd<0>
"SS of RER depr., ann. QQ %" ss_dl_z_tnd<0>
"SS of risk premium, %" ss_prem<2>
"SS of lending premium, %" ss_prem_d<8.5>

"SS of chng. of vel., ann. QQ %" ss_dl_v<-2.02027>

% -----
% ----- Cyclical coefficients -----
% ----- Fiscal parameters -----

"Fiscal, grants. persist." d7<0.5>
" grev_y_cyc response to cons * rev/GDP" v1<0.012>
" grev_y_cyc response to imports * rev/GDP" v2<0.006>
" grev_y_cyc response outputgap * rev/GDP" v3<0.02>
" grev_y_str, lag" v4<0.99>

"oexp_y_cyc,lag %" u1<0>
"oexp_y_cyc response output gap * oexp/GDP" u2<0.012>
"oexp_y_str, lag %" u3<0.95>

"gdem_y_cyc (Govt demand G&S), lag" t1<0>
"gdem_y_cyc response outputgap * gdem/GDP" t2<0.023>
"gdem_y_str (Govt demand G&S), lag" t3<0.95>
"gdem_y_discr (Govt demand G&S), lag" t4<0.7>
"gdem_y_discr in fiscal impulse" s1<1>

```

```

% ----- GDP components -----
"Consumption gap, lag" a1_cons<0.65>
"Consumption gap, expect." a2_cons<0>
"Consumption gap, RMCi" a3_cons<0.1>
"Consumption gap, output gap" a4_cons<0.2>
"Consumption gap, fisc. imp." a5_cons<0.25>
"Consumption gap, domestic in RMCi" a6_cons<1>

"Investment gap, lag" a1_inv<0.65>
"Investment gap, expect." a2_inv<0>
"Investment gap, RMCi(RIR, RER)" a3_inv<0.1>
"Investment gap, output gap" a4_inv<0.2>
"Investment gap, fisc. imp." a5_inv<0.15>

"Investment gap, domestic in RMCi" a6_inv<1>
"Export gap, lag" a1_exp<0.65>
"Export gap, expect." a2_exp<0>
"Export gap, RMCi(RIR, REER)" a3_exp<0.2>
"Export gap, foreign dem." a5_exp<0.3>
"Export gap, domestic in RMCi" a6_exp<0>
"Import gap, REER coefficient" a1_imp<0.3>
"Agric.output gap AR(1)" r1_y_agr<0.4>

% ----- Phillips curves -----

"Core Phillips curve, lag" b1<0.45>
"Core Phillips curve, RMC" b2<0.2>
"Core Phillips curve, direct imp." b3<0.05>
"Core Phillips curve, domestic in RMC" b4<0.8>

"Food Phillips c., lag" bf1<0.35>
"Food Phillips c.,RMC indirect importprice" bf2<0.1>
"Food Phillips c.,direct importprice" bf3<0.1>
"+Food Phillips c., agric.output gap neg." bf4<1.5>

"Energy Phillips curve, lag" be1<0.25>
"Energy Phillips curve, RMC" be2<0.04>
"Energy Phillips curve, direct imp." be3<0.02>

```

% ----- Monetary policy rule -----

"Policy rule, lag" c1<0.8>
"Policy rule, inflation" c2<0.5>
"Policy rule, output gap" c3<0.5>
"Policy rule, FX target" c4<0>
"Inflation target, persist." c5<0.9>
"Lending premium, lag" c6<0.9>

% ----- Exchange rate -----

"Exchng rate, UIP" e1<0.2>
"Exchng rate expect., forward" e2<0.2>
"Premium, persist." e3<0.9>
"Exchng rate target, persist." e4<0>
"Exchng rate target, infl. dev." e5<0.3>
"Exchng rate target, REER gap" e6<0.85>

% ----- Money demand -----

"Real money dem., lag" m1<0.7>
"Real money dem., interest rate" m2<0.5>
"Change velocity, lag" m3<0.9>

% -----

% ----- Weights -----

"Import gap, cons.share imports" w_imp_cons<0.445714>
"Import gap, inv.share imports" w_imp_inv<0.156>
"Import gap, govdemand share imports" w_imp_gdem<0.276>
"Import gap, exp.share imports" w_imp_exp<0.122286>

"GDP gap, cons. share GDP" w_y_cons<0.78>
"GDP gap, inv. share GDP" w_y_inv<0.13>
"GDP gap, govdemand share GDP" w_y_gdem<0.23>
"GDP gap, exp. share GDP" w_y_exp<0.21>
"GDP gap, imp. share GDP" w_y_imp<0.35>

"Weight of core in CPI" w_core<0.7747>
"Weight of food in CPI" w_food<0.1577>
"Weight of energy in CPI" w_ener<0.0676>

% -----

% ----- Trend persistences -----

"Cons. tnd. persist." r_cons<0.95>
"Inv. tnd. persist." r_inv<0.95>
"Gov. demand, tnd. persist." r_gdem<0.95>
"Export tnd. persist." r_exp<0.95>
"Import tnd. persist." r_imp<0.95>
"Agric.output tnd. persist." r2_y_agr<0.8>
"RER tnd. persist." r_z<0.95>

"Food rel. price tnd. persist." r_rp_food<0.9>
"Energy rel. price tnd. persist." r_rp_ener<0.9>

% -----

% ----- Foreign block parameters -----

"SS of foreign CPI, ann. QQ %" ss_dl_cpistar<1.98026>
"SS of foreign real intr. rate, %" ss_rstar_tnd<0>
"SS of foreign rel. food price, ann. QQ %" ss_dl_rp_foodstar_tnd<0>
"SS of foreign rel. ener. price, ann. QQ %" ss_dl_rp_enerstar_tnd<0>
"foreign demand persist." r_ystar<0.94>
"foreign CPI persist." r_cpistar<0.8>
"foreign interest rate persist." r_istar<0.95>
"for. interest rate tnd. persist." r_rstar_tnd<0.9>
"for. rp food persist." r_rp_foodstar_gap<0.62>
"for. rp energy persist." r_rp_enerstar_gap<0.73>
"for. rp food tnd. persist." r_rp_foodstar_tnd<0.9>
"for. rp energy tnd. persist." r_rp_enerstar_tnd<0.9>

```

% -----
% ----- Transition equations -----
% -----
!transition_equations

% -----
% ----- Fiscal policy -----
grants_y = d7<0.5> * grants_y{-1} + (1-d7<0.5>) * ss_grants_y<5> + shock_grants_y<σ=2.5>;

% tax revenue -----
grev_y = grev_y_str + grev_y_cyc + grev_y_discr;
grev_y_cyc = v1<0.012> * l_cons_gap + v2<0.006> * l_imp_gap + v3<0.02> * l_y_gap +
shock_grev_y_cyc<σ=0>;
grev_y_str = + v4<0.99> * grev_y_str{-1} + (1-v4<0.99>) * ss_grev_y_str<21> + shock_grev_y_str<σ=0.1>;
grev_y_discr = shock_grev_y_discr<σ=1.7321>;

% other expenditure -----
oexp_y = oexp_y_str + oexp_y_cyc + oexp_y_discr;

oexp_y_cyc = u1<0> * oexp_y_cyc{-1} - u2<0.012> * l_y_gap + shock_oexp_y_cyc<σ=0>;
oexp_y_str = + u3<0.95> * oexp_y_str{-1} + (1-u3<0.95>) * ss_oexp_y_str<6> +
shock_oexp_y_str<σ=0.1732>;
oexp_y_discr = shock_oexp_y_discr<σ=1.7321>;

% Govt. demand-----
gdem_y = gdem_y_str + gdem_y_cyc + gdem_y_discr;
gdem_y_cyc = t1<0> * gdem_y_cyc{-1} - t2<0.023> * l_y_gap + shock_gdem_y_cyc<σ=0>;
gdem_y_str = + t3<0.95> * gdem_y_str{-1} + (1-t3<0.95>) * ss_gdem_y_str<26> +
shock_gdem_y_str<σ=0.075>;
gdem_y_discr = t4<0.7> * gdem_y_discr{-1} + shock_gdem_y_discr<σ=1.7321>;
l_gdem/100 = log(gdem_y / 100) + l_y/100;
l_gdem_tnd/100 = log(gdem_y_str / 100) + l_y_tnd/100;

% budget deficit -----
def_y = gdem_y + oexp_y - grev_y;
def_y_str = gdem_y_str + oexp_y_str - grev_y_str;
def_y_cyc = gdem_y_cyc + oexp_y_cyc - grev_y_cyc;
def_y_discr = gdem_y_discr + oexp_y_discr - grev_y_discr;

% fiscal impulse -----
fisc_imp = + s1<1> * gdem_y_discr + oexp_y_discr - grev_y_discr + def_y_str - def_y_str{-1};

```

```

% -----
% ----- Real demand -----
% ----- Consumption gap -----
% (note fisc_imp is indirect effect of l_gdem_gap)

l_cons_gap = ...
    + a1_cons(0.65) * l_cons_gap{-1} ...
    + a2_cons(0) * e_l_cons_gap ...
    - a3_cons(0.1) * rmci_cons ...
    + a4_cons(0.2) * l_y_gap ...
    + a5_cons(0.25) * fisc_imp ...
    + shock_l_cons_gap( $\sigma=2.5$ );

rmci_cons = a6_cons(1) * r4_gap + (1 - a6_cons(1)) * -l_z_gap;

e_l_cons_gap = l_cons_gap{+1};

% ----- Investment gap -----
% (note fisc_imp is indirect effect of l_gdem_gap)

l_inv_gap = ...
    + a1_inv(0.65) * l_inv_gap{-1} ...
    + a2_inv(0) * e_l_inv_gap ...
    - a3_inv(0.1) * rmci_inv ...
    + a4_inv(0.2) * l_y_gap ...
    + a5_inv(0.15) * fisc_imp ...
    + shock_l_inv_gap( $\sigma=6$ );

rmci_inv = a6_inv(1) * r4_gap + (1 - a6_inv(1)) * -l_z_gap;

e_l_inv_gap = l_inv_gap{+1};

% ----- Export gap -----

l_exp_gap = ...
    + a1_exp(0.65) * l_exp_gap{-1} ...
    + a2_exp(0) * e_l_exp_gap ...
    - a3_exp(0.2) * rmci_exp ...
    + a5_exp(0.3) * l_ystar_gap ...
    + shock_l_exp_gap( $\sigma=6$ );

rmci_exp = a6_exp(0) * r4_gap + (1 - a6_exp(0)) * -l_z_gap;
e_l_exp_gap = l_exp_gap{+1};

```



```

% ----- Import gap -----
% (REER added same for all import demands, coeff REER>0, so -a1_imp)

l_imp_gap = ...
    + w_imp_cons(0.445714) * l_cons_gap ...
    + w_imp_inv(0.156) * l_inv_gap ...
    + w_imp_gdem(0.276) * l_gdem_gap ...
    + w_imp_exp(0.122286) * l_exp_gap ...
    - a1_imp(0.3) * l_z_gap ...
    + shock_l_imp_gap( $\sigma=0.5$ );

% ----- Output gap -----
% (note direct effect of l_gdem_gap on l_y_gap)

l_y_gap = ...
    + w_y_cons(0.78) * l_cons_gap ...
    + w_y_inv(0.13) * l_inv_gap ...
    + w_y_gdem(0.23) * l_gdem_gap ...
    + w_y_exp(0.21) * l_exp_gap ...
    - w_y_imp(0.35) * l_imp_gap ...
    + shock_l_y_gap( $\sigma=0.0001$ );
e_l_y_gap = l_y_gap{+1};

% -----
% ----- Agricultural output gap -----

l_y_agr_gap = r1_y_agr(0.4) * l_y_agr_gap{-1} + shock_l_y_agr_gap( $\sigma=0.3$ );
e_l_y_agr_gap = l_y_agr_gap{+1};

% ----- Prices -----
% ----- Core Phillips curve -----

dl_cpi_core = ...
    + b1(0.45) * dl_cpi_core{-1} ...
    + (1 - b1(0.45) - b3(0.05)) * (e_dl_cpi_core) ...
    + b3(0.05) * dl_cpi_core_direct ...
    + b2(0.2) * rmc ...
    + shock_dl_cpi_core( $\sigma=2$ );

rmc = b4(0.8) * l_y_gap + (1 - b4(0.8)) * l_z_gap;
dl_cpi_core_direct = dl_cpistar + dl_s - dl_z_tnd;
e_dl_cpi_core = dl_cpi_core{+1};

l_rp_cpi_core = l_cpi_core - l_cpi;

```


% ----- Food Phillips curve -----

% agric. output gap in food inflation, with coeff as BNR-IMF -0.5

```
dl_cpi_food = ...
    + bf1<0.35> * dl_cpi_food{-1} ...
    + (1 - bf1<0.35> - bf3<0.1>) * (e_dl_cpi_food) ...
    + bf3<0.1> * dl_cpi_food_direct ...
    + bf2<0.1> * (l_rp_foodstar_gap + l_z_gap + l_rp_cpi_core_gap - l_rp_cpi_food_gap) ...
    - bf4<1.5> * l_y_agr_gap ...
    + shock_dl_cpi_food<σ=9>;
dl_cpi_food_direct = dl_foodstar - dl_rp_foodstar_tnd + dl_s - dl_z_tnd ...
    - dl_rp_cpi_core_tnd + dl_rp_cpi_food_tnd;
e_dl_cpi_food = dl_cpi_food{+1};
l_rp_cpi_food = l_cpi_food - l_cpi;
```

% ----- Energy Phillips curve -----

```
dl_cpi_ener = ...
    + be1<0.25> * dl_cpi_ener{-1} ...
    + (1 - be1<0.25> - be3<0.02>) * (e_dl_cpi_ener)...
    + be3<0.02> * dl_cpi_ener_direct ...
    + be2<0.04> * (l_rp_enerstar_gap + l_z_gap + l_rp_cpi_core_gap - l_rp_cpi_ener_gap)
    + shock_dl_cpi_ener<σ=7.1>;
dl_cpi_ener_direct = dl_enerstar - dl_rp_enerstar_tnd + dl_s - dl_z_tnd ...
    - dl_rp_cpi_core_tnd + dl_rp_cpi_ener_tnd;
e_dl_cpi_ener = dl_cpi_ener{+1};

l_rp_cpi_ener = l_cpi_ener - l_cpi;
```

% ----- Headline CPI level -----

```
l_cpi = ...
    +w_core<0.7747> * l_cpi_core
    + w_food<0.1577> * l_cpi_food
    + w_ener<0.0676> * l_cpi_ener
    + shock_l_cpi<σ=0.0001>;
e_dl_cpi = dl_cpi{+1};

0 = w_core<0.7747> * l_rp_cpi_core_gap + w_food<0.1577> * l_rp_cpi_food_gap + w_ener<0.0676> ....
    * l_rp_cpi_ener_gap;
```

% -----

% ----- Monetary policy -----

% (standard, except ER deviation, but coeff=0)

% ----- Inflation forecast based rule -----

```
i = c1<0.8> * i{-1} + (1 - c1<0.8>) * (...  
    + i_tnd ...  
    + c2<0.5> * d4l_cpi_dev ...  
    + c3<0.5> * l_y_gap ...  
    + c4<0> * (dl_s - dl_s_tar) ...  
    + shock_i<σ=1>);
```

```
d4l_cpi_dev = d4l_cpi{+4} - d4l_cpi_tar;
```

% nominal interest trend/ neutral level

```
i_tnd = r_tnd + d4l_cpi_tar + dl_rp_cpi_core_tnd;
```

% ----- Real interest rate -----

```
r = i - e_dl_cpi_core;
```

```
r_tnd = rstar_tnd + prem + e_dl_z_tnd;
```

```
r4_gap = (r_gap + r_gap{+1} + r_gap{+2} + r_gap{+3}) / 4 + prem_d_gap;
```

```
e4_r_gap = (r_gap + r_gap{+1} + r_gap{+2} + r_gap{+3}) / 4;
```

% ----- Lending premium -----

```
prem_d = prem_d_gap + ss_prem_d<8.5>;
```

```
prem_d_gap = c6<0.9> * prem_d_gap{-1} + shock_prem_d_gap<σ=0.8>;
```

% ----- Inflation target -----

```
d4l_cpi_tar = c5<0.9> * d4l_cpi_tar{-1} + (1 - c5<0.9>) * ss_d4l_cpi_tar<4.87902> + shock_d4l_cpi_tar<σ=0.1>;
```

```
d4l_cpi_core_tar = d4l_cpi_tar + d4l_rp_cpi_core_tnd;
```

```
d4l_cpi_food_tar = d4l_cpi_tar + d4l_rp_cpi_food_tnd;
```

```
d4l_cpi_ener_tar = d4l_cpi_tar + d4l_rp_cpi_ener_tnd;
```

```

% -----
% ----- Exchange rate -----
% (UIP & move to ER-target, det. by RER-tnd & target infl.diff.)

l_s = ...
    + e1<0.2> * (e_l_s - (i - istar - prem)/4) ...
    + (1 - e1<0.2>) * (l_s{-1} + dl_s_tar/4) ...
    + shock_l_s<σ=0.35>;
e_l_s = ...
    + e2<0.2> * l_s{+1} ...
    + (1 - e2<0.2>) * (l_s{-1} + 2*(dl_z_tnd + d4l_cpi_tar + dl_rp_cpi_core_tnd -
ss_dl_cpistar<1.98026>)/4);
prem = e3<0.9> * prem{-1} + (1 - e3<0.9>) * ss_prem<2> + shock_prem<σ=0.1>;

dl_s_tar = e4<0> * dl_s_tar{-1} + (1 - e4<0>)...
    * ( dl_z_tnd + d4l_cpi_tar + dl_rp_cpi_core_tnd - ss_dl_cpistar<1.98026>
    - e5<0.3> * d4l_cpi_dev - e6<0.85> * l_z_gap ...
    + shock_dl_s_tar<σ=0.3>;
l_z = l_s + l_cpistar - l_cpi_core;
e_dl_z = dl_z{+1};
e_dl_z_tnd = dl_z_tnd{+1};

% -----
% ----- Real money demand -----
% (function of nominal i dev., cp. BNR: change in r-tnd)

l_rmd = l_md - l_cpi;
dl_rmd = ...
    + m1<0.7> * dl_rmd{-1} ...
    + (1 - m1<0.7>) * (dl_y - dl_v + m2<0.5> * (i - i_tnd)) ...
    + shock_dl_rmd<σ=1>;
dl_v = m3<0.9> * dl_v{-1} + (1 - m3<0.9>) * ss_dl_v<-2.02027> + shock_dl_v<σ=0.1>;

```

```

% -----
% ----- Trends -----
% (weights for output-tnd are same as for output-gap)

dl_cons_tnd = r_cons<0.95> * dl_cons_tnd{-1} + (1 - r_cons<0.95>) * ss_dl_y_tnd<7.23207> +
shock_dl_cons_tnd< $\sigma=0.25$ >;
dl_inv_tnd = r_inv<0.95> * dl_inv_tnd{-1} + (1 - r_inv<0.95>) * ss_dl_y_tnd<7.23207> +
shock_dl_inv_tnd< $\sigma=0.5$ >;
dl_exp_tnd = r_exp<0.95> * dl_exp_tnd{-1} + (1 - r_exp<0.95>) * ss_dl_y_tnd<7.23207> +
shock_dl_exp_tnd< $\sigma=0.75$ >;
dl_imp_tnd = r_imp<0.95> * dl_imp_tnd{-1} + (1 - r_imp<0.95>) * ss_dl_y_tnd<7.23207> +
shock_dl_imp_tnd< $\sigma=0.25$ >;

dl_y_tnd = ...
    + w_y_cons<0.78> * dl_cons_tnd ...
    + w_y_inv<0.13> * dl_inv_tnd ...
    + w_y_gdem<0.23> * dl_gdem_tnd ...
    + w_y_exp<0.21> * dl_exp_tnd ...
    - w_y_imp<0.35> * dl_imp_tnd;
dl_y_agr_tnd = r2_y_agr<0.8> * dl_y_agr_tnd{-1} + (1 - r2_y_agr<0.8>)...
    * ss_dl_y_agr_tnd<7.23207> + shock_dl_y_agr_tnd< $\sigma=0.06$ >; % no role yet
dl_z_tnd = r_z<0.95> * dl_z_tnd{-1} + (1 - r_z<0.95>) * ss_dl_z_tnd<0> + shock_dl_z_tnd< $\sigma=0.1$ >;

dl_rp_cpi_food_tnd = r_rp_food<0.9> * dl_rp_cpi_food_tnd{-1} + (1 - r_rp_food<0.9>) *
ss_dl_rp_cpi_food_tnd<1.98026>
    + shock_dl_rp_cpi_food_tnd
dl_rp_cpi_ener_tnd = r_rp_ener<0.9> * dl_rp_cpi_ener_tnd{-1} + (1 - r_rp_ener<0.9>) *
ss_dl_rp_cpi_ener_tnd<0>
    + shock_dl_rp_cpi_ener_tnd< $\sigma=831$ >;

```

```

% -----
% ----- External sector -----

l_ystar_gap = r_ystar(0.94) * l_ystar_gap{-1} + shock_l_ystar_gap(σ=0.45);

dl_cpistar = r_cpistar(0.8) * dl_cpistar{-1} + (1 - r_cpistar(0.8)) * ss_dl_cpistar(1.98026) + shock_dl_cpistar(σ=3.8);
e_dl_cpistar = dl_cpistar{+1};

istar = r_istar(0.95) * istar{-1} + (1 - r_istar(0.95)) * (ss_rstar_tnd(0) + ss_dl_cpistar(1.98026)) + shock_istar(σ=0.45);

rstar_tnd = r_rstar_tnd(0.9) * rstar_tnd{-1} + (1 - r_rstar_tnd(0.9)) * ss_rstar_tnd(0) + shock_rstar_tnd(σ=0.5);

l_rp_foodstar = l_foodstar - l_cpistar;
l_rp_enerstar = l_enerstar - l_cpistar;
l_rp_foodstar_gap = r_rp_foodstar_gap(0.62) * l_rp_foodstar_gap{-1} + shock_l_rp_foodstar_gap(σ=5.3);
l_rp_enerstar_gap = r_rp_enerstar_gap(0.73) * l_rp_enerstar_gap{-1} + shock_l_rp_enerstar_gap(σ=13.9);
dl_rp_foodstar_tnd = r_rp_foodstar_tnd(0.9) * dl_rp_foodstar_tnd{-1} + (1 - r_rp_foodstar_tnd(0.9)) ....
                    * ss_dl_rp_foodstar_tnd(0) + shock_dl_rp_foodstar_tnd
dl_rp_enerstar_tnd = r_rp_enerstar_tnd(0.9) * dl_rp_enerstar_tnd{-1} + (1 - r_rp_enerstar_tnd(0.9))
                    * ss_dl_rp_enerstar_tnd(0) + shock_dl_rp_enerstar_tnd851:

% -----
% ----- Trend/gap identities -----
!for
    l_y, l_y_agr, l_cons, l_inv, l_exp, l_imp, l_gdem,
    r, l_z
    l_rp_cpi_food, l_rp_cpi_ener, l_rp_cpi_core
    l_rp_foodstar, l_rp_enerstar
!do
!transition_equations
    ? = ?_tnd + ?_gap;
!end

% ----- Growth rates -----

!for
    y, y_tnd, cons, cons_tnd, inv, inv_tnd, gdem, gdem_tnd, exp, exp_tnd, imp, imp_tnd
    y_agr, y_agr_tnd,
    cpi, cpi_core, cpi_food, cpi_ener
    p_cpi_food, rp_cpi_ener, rp_cpi_core, rp_cpi_food_tnd, rp_cpi_ener_tnd, rp_cpi_core_tnd
    s, z, z_tnd, rmd
    cpistar, foodstar, enerstar
    rp_foodstar, rp_enerstar, rp_foodstar_tnd, rp_enerstar_tnd
!do
!transition_equations
    dl_? = 4 * (l_? - l_?{-1});
    d4l_? = (l_? - l_?{-4});
!end

```

```

% -----
% ----- Measurement variables/equations -----
% -----
% -----
% ----- Hard observations -----

!for
    def_y, grants_y, grev_y
    l_y, l_y_agr, l_cons, l_gdem, l_inv, l_exp, l_imp
    l_cpi, l_cpi_core, l_cpi_food, l_cpi_ener
    i, prem_d, l_s, l_md
    l_ystar_gap, l_cpistar, istar, rstar_tnd
    l_foodstar, l_enerstar
    l_rp_foodstar_gap, l_rp_enerstar_gap
!do
!measurement_variables
    obs_?
!measurement_equations
    obs_? = ?;
!end

% -----
% ----- Expert/judgemental tunes -----AK added Oct 13, def_y, grev_y

!for
    gdem_y, oexp_y, def_y, grev_y
    gdem_y_str, oexp_y_str, grev_y_str
    gdem_y_cyc, oexp_y_cyc, grev_y_cyc
    grev_y_discr, oexp_y_discr, gdem_y_discr
    l_y_tnd, l_cons_tnd, l_inv_tnd, l_gdem_tnd, l_exp_tnd, l_imp_tnd
    d4l_cpi_tar, l_rp_cpi_food_tnd, l_rp_cpi_ener_tnd
    r_tnd, l_z_tnd, prem, dl_s_tar
    def_y_str, def_y_discr, grants_y
    l_y_gap, l_cons_gap, l_inv_gap, l_gdem_gap, l_exp_gap, l_imp_gap
    l_y_agr_gap
    l_z_gap
    l_y, d4l_y, dl_s
!do
!measurement_variables
    tune_?
!measurement_equations
    tune_? = ?;
!end

```

```

% -----
% ----- Auxiliary variables/equations for shock tuning in the filter -----

!for
    shock_dl_cons_tnd( $\sigma=0.25$ )
    shock_dl_cpi_ener( $\sigma=7.1$ )
    shock_dl_cpi_core( $\sigma=2$ )
!do
!transition_variables
aux_?
!transition_equations
aux_? = ?;
!measurement_variables
tune_?
!measurement_equations
tune_? = aux_?;
!end

% -----
% ----- Reporting equations -----
% -----

!parameters
"Share of import prices in CPI" mu_pimp( $0.35$ )
"Share of export prices in GDP deflator" mu_pexp( $0.2$ )
"share of fcy debt in total debt" r_debt_fcy_rat( $0.9$ )
"interest rate fcy debt persistence" r_debt_fcy_intrate_pers( $0.9$ )
"interest rate lcy debt persistence" r_debt_lcy_intrate_pers( $0.9$ )
gamma_r( $0.5$ ); gamma_k( $0.5$ ); gamma_BP_tnd( $0.9$ );

% AK 7/14/23 replace k_bar by l_BP_tnd; % log(BP)=k, BP cum inflows mln$,
% BP does not go to fixed ss level k_bar=log(6500), but to trend
% with initial value trend set equal to actual in readData

"Import share in consumption" lam_imp_cons( $0.2$ )
"Import share in investment" lam_imp_inv( $0.42$ )
"Import share in govdemand" lam_imp_gdem( $0.42$ )
"Import share in exports" lam_imp_exp( $0.20381$ )

```


!reporting_equations

% True percentages (added Oct 28, '22)

!for

y, cons, inv, gdem, exp, imp
cpi, cpi_core, cpi_food, cpi_ener
s, z

!do

pct_? = exp(dl_?/100)*100 - 100;
pct4_? = exp(d4l_?/100)*100 - 100;

!end

!for

i, r

!do

pct_? = exp(?/100)*100 - 100;

!end

% Levels exchange rate and nominal money ds = exp(l_s/100);

md = exp(l_md/100);

dl_md = 4*(l_md - l_md{-1}); % ak 7/15/23 must[dbFcats be annualized

% Deflators (recall: all annualized, *4)

dl_pexp = dl_pexpstardl_pdom = (dl_cpi - mu_pimp(0.35) * dl_pimp) / (1 - mu_pimp(0.35));

dl_py = mu_pexp(0.2) * dl_pexp + (1 - mu_pexp(0.2)) * dl_pdom;

% deflator for govt demand assumed equal to dl_cpi_core to simplify

% Nominal growth rates (recall:all annualized, *4)

dl_nexp = dl_exp + dl_pexp;

dl_nimp = dl_imp + dl_pimp;

l_ny = dl_y + dl_py;

dl_ngdem = dl_gdem + dl_cpi_core;

dl_ncons = dl_cons + dl_cpi_core; % 4/19 AK for ratios (% to nom GDP)

dl_ninv = dl_inv + dl_cpi_core;

% Nominal levels exports, imports, GDP

nexp = exp(log(nexp{-1}) + dl_nexp/100/4);

nimp = exp(log(nimp{-1}) + dl_nimp/100/4);

ny = exp(log(ny{-1}) + dl_ny/100/4); % get ny{-1} etc from dbAUX to have start value forecast

ngdem = exp(log(ngdem{-1}) + (dl_ngdem/100)/4); % idem

ncons = exp(log(ncons{-1}) + dl_ncons/100/4); % 4/19/23 AK for ratios

ninv = exp(log(ninv{-1}) + dl_ninv/100/4);

tb_rat = (nexp - nimp) / ny *100;

% Capital account (Ostry/Ghosh/Chamon,2012:log(BP)=k; define rstar here, not done in model)

% 7/14/23 trend eq for ss cumul capital inflows in line with real GDP, instead of k_bar

rstar = istar - e_dl_cpistar;

% first evaluate trend

dl_BP_tnd = gamma_BP_tnd(0.9) * dl_BP_tnd{-1} + (1-gamma_BP_tnd(0.9)) * ss_dl_y_tnd(7.23207);

l_BP_tnd = dl_BP_tnd/4 + l_BP_tnd{-1};

% then evaluate level

l_BP = l_BP{-1} + gamma_r(0.5) * ((r - rstar - prem)/4 - e_dl_z) ...

- gamma_k(0.5) * (l_BP{-1} - l_BP_tnd);

dBp_usd = exp(l_BP/100) - exp(l_BP{-1}/100);

% evaluate debt and deficit first before rest of BOP and MON

% ak debt changes with deficit-/grants, nb def_y and grants are over Q-GDP, as is debt

debt_y = ...

+ debt_lcy_y{-1} * (1) / exp(dl_ny/400) ...

+ debt_fcy_y{-1} * exp(dl_s/400) / exp(dl_ny/400) ...

+ def_y - grants_y;

debt_fcy_rat = r_debt_fcy_rat(0.9) * debt_fcy_y{-1} / debt_y{-1} + (1 - r_debt_fcy_rat(0.9)) * ss_debt_fcy_rat(0.846154);

debt_fcy_y = debt_fcy_rat * debt_y;

debt_lcy_y = debt_y - debt_fcy_y; % residual

def_lcy_y = debt_lcy_y - debt_lcy_y{-1} * (1) / exp(dl_ny/400);

def_fcy_y = debt_fcy_y - debt_fcy_y{-1} * exp(dl_s/400) / exp(dl_ny/400);

% ak we need interest-rate update equations:

% i_debt_fcy = r_debt_fcy_intrate_pers * i_debt_fcy{-1} + (1 -

% r_debt_fcy_intrate_pers) * (ss_rstar_tnd + ss_dl_cpistar + ss_prem_debt_fcy)

% i_debt_lcy = r_debt_lcy_intrate_pers * i_debt_lcy{-1} + (1 -

% r_debt_lcy_intrate_pers) * (i_tnd + ss_prem_debt_lcy)

intp_fcy_y = i_debt_fcy{-1}/400 * debt_fcy_y{-1} * exp(dl_s/400) / exp(dl_ny/400);

intp_lcy_y = i_debt_lcy{-1}/400 * debt_lcy_y{-1} * (1) / exp(dl_ny/400);

% External flow equilibrium (current account (RM=E-M) converted to mln \$)

% still need evaluate levels govt interest payments in usd for BOP

DBG_usd = (((def_fcy_y/100) * ny) / s) * 1000; % for bor deficit RWFbln then to \$mln

NFG_usd = (((grants_y/100) * ny) / s) * 1000; % net foreign grants govt RWFbln then to \$mln

dNFA_usd = (nexp - nimp) / s * 1000 + NFG_usd + DBG_usd + dBp_usd;

% ak 7/15/23 trace monetary flows, share of bank in domestic financing deficit

def_lcy_bank_y = 0.5 * def_lcy_y; % nb still over Q GDP in %!

NCG = NCG{-1} + (def_lcy_bank_y/100) * ny;

NFA = NFA{-1} + dNFA_usd * s/1000; % ignores ER valuation changes

% Monetary flow equilibrium in bln RWF, private credit residual

dNCP = diff(md) - dNFA_usd * s / 1000 - diff(NCG);